P07

Variational Principles and Minimum Dissipation in Heat Conduction

Authors	L. Onsager
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Abstract

Reciprocal relations in irreversible processes imply that steady-state heat conduction in linear conductive networks minimizes the total dissipation rate, mirroring Rayleigh's principle of least dissipation of energy.

Abstract quoted from L. Onsager et al., Physical Review (1931).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?